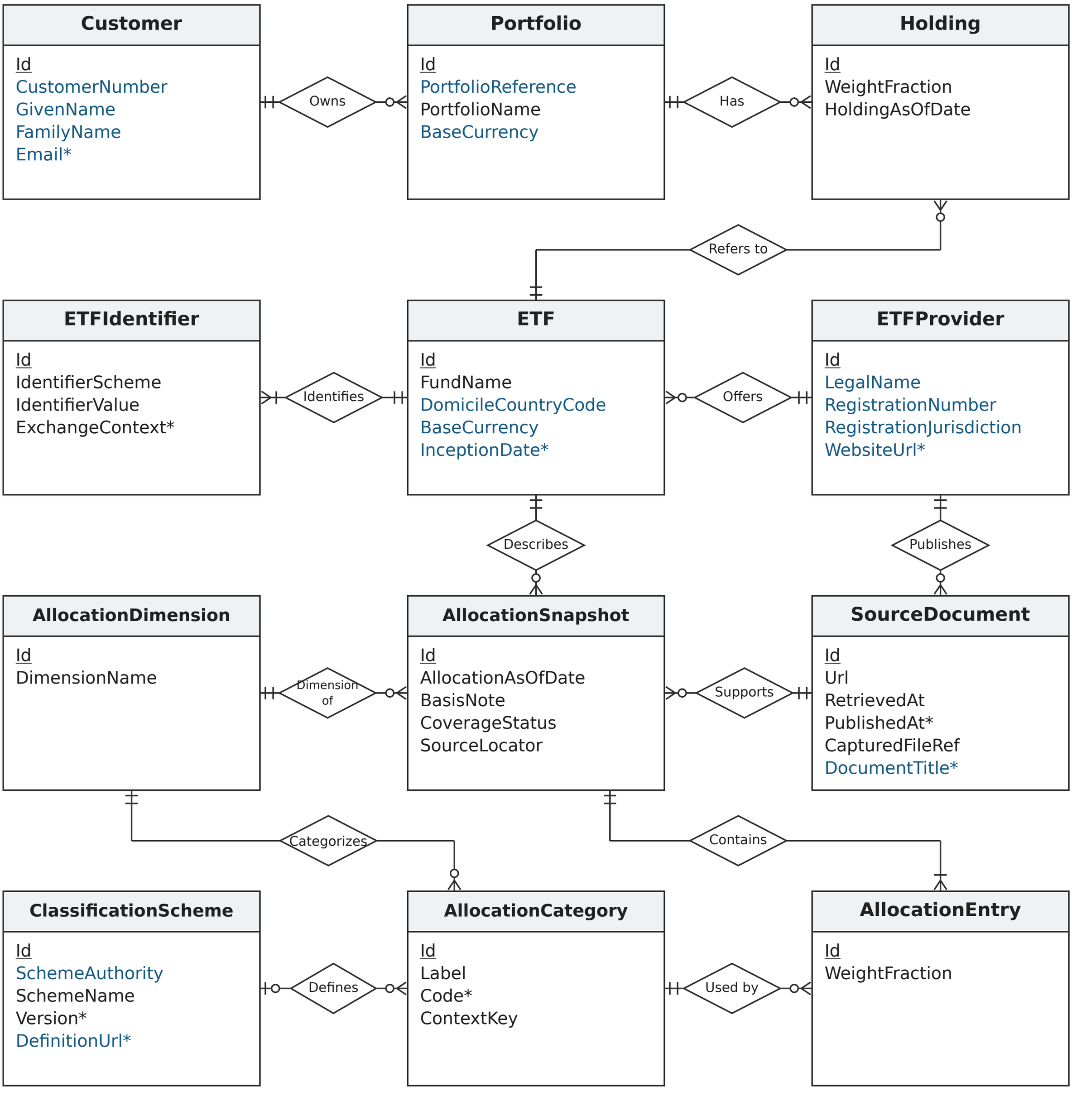
Dated ETF allocation disclosures

CSC501 Milestone 4 preparation | Revised conceptual ER diagram | 30 October 2026

Compact attributes; relationship diamonds and crow's-foot endpoints. Identity rules are on page 2.



How to read the endpoints

Exactly one — Zero or one — One or more — Zero or more

Underlined Id = internal record identifier, not a foreign key. Business identity rules are specified on page 2.

* = optional attribute. Other attributes are required. Blue attributes are additions or clarified descriptions.

Scope: Customer is an individual at one bank; ETFProvider is a registered company; ETF is one fund/share class.

Holding stores portfolio-ETF weight and date; AllocationEntry stores snapshot-category weight.

Holding and disclosure dates differ. A source is a captured version; category and snapshot dimensions must agree.

A portfolio has one owner; an ETF has one sampled provider; a snapshot has one primary captured source.

Business identity and implementation rules

Read with page 1. Parentheses indicate composite uniqueness rules. Related entities below denote relationships. The rules make prototype scope decisions explicit.

Customer

Business key: CustomerNumber, within the single bank.

CustomerNumber is the bank-issued customer reference.

GivenName and FamilyName describe the individual.

Email is optional and is not assumed unique.

ETFProvider

Business key: (RegistrationJurisdiction,

RegistrationNumber). LegalName describes the company.

The jurisdiction identifies the issuing registry.

Use verified registry values or labeled synthetic data.

Portfolio

Business key: (owning Customer, PortfolioReference).

Reference is an existing bank/customer portfolio reference.

PortfolioName is descriptive, not a key.

BaseCurrency records the reporting currency.

ETF

Business identity is supplied by its ETFIdentifier records.

FundName is descriptive and is not assumed unique.

Scope: one fund/share class, with one sampled provider.

DomicileCountryCode and BaseCurrency describe it.

Holding

Business key: (Portfolio, ETF, HoldingAsOfDate).

One aggregate position per ETF, portfolio and date.

WeightFraction is in [0, 1]; it is an association attribute.

The holding date is independent of disclosure dates.

ETFIdentifier

ISIN uniqueness: (IdentifierScheme, IdentifierValue).

Ticker uniqueness: (IdentifierScheme, IdentifierValue,

ExchangeContext). A ticker requires an exchange.

ExchangeContext is absent for ISIN in this prototype.

AllocationDimension

Business key: DimensionName.

Allowed values: Sector and Geography.

These are values, rather than part of an attribute name.

Both categories and snapshots belong to one dimension.

ClassificationScheme

Known-version uniqueness: (SchemeAuthority,

SchemeName, Version), when Version is known.

An unknown version remains unknown; Id identifies

that record. Do not invent a business key for it.

AllocationSnapshot

Scoped uniqueness: (ETF, AllocationDimension,

AllocationAsOfDate, SourceDocument, SourceLocator).

One dated allocation block per such source location.

CoverageStatus: Complete, Partial or Unknown.

SourceDocument

Business key: (Url, RetrievedAt). Store time in UTC.

Each record is one capture of a URL at a retrieval time.

CapturedFileRef resolves to the retained captured file.

Url alone cannot identify a historical source version.

AllocationCategory

Scoped uniqueness: (AllocationDimension, ContextKey,

Label). Assume labels are unique inside that context.

ContextKey identifies a verified scheme/version or a

source capture + allocation block. Code is optional.

AllocationEntry

Business key: (AllocationSnapshot, AllocationCategory).

One weight per category in a snapshot; weight in [0, 1].

Its category and snapshot dimensions must agree.

A recorded disclosure has at least one entry.

Unknown or unverified schemes/versions: use disclosure-local category contexts; similar labels do not prove equivalence.

Fractions use 0.30 for 30%. Complete-disclosure totals need a documented rounding tolerance; partial totals need not be 1.

SQL mapping: every relation has an autogenerated integer Id. Map relationships to FKs and enforce the scoped rules above.

Optional-key rules need null-aware uniqueness. Minimum one entry per completed snapshot needs transactional validation.

Revision prepared with AI assistance. Team members must review, validate and document their evaluation before submission.